

Supplementary Material

S3 — TRIPOD Checklist (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis Or Diagnosis)

Paper Title: *Explainable Machine Learning for Student Burnout Classification and Risk Stratification: A Mixed-Methods Study with Qualitative Triangulation*

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22-item checklist for prediction model development studies — Collins, Reitsma, Altman & Moons (2015)

Model Type: Classification (Prediction Model Development — 10-Fold Stratified Cross-Validation)

Outcome: Binary classification of High Academic Burnout (burnout_score = 3) vs. Low/Medium Burnout (burnout_score = 1 or 2) in Bangladeshi university students.

TRIPOD Checklist

Section	Item	Checklist Item	Manuscript Location	Reported?
Title and Abstract	1	Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted.	Title, Abstract (Methods paragraph)	Yes
Introduction	2a	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models.	Section 1.1, Section 2.4, Table 1	Yes
	2b	Specify the objectives, including whether the study describes the development or validation of the model or both.	Section 1.7, Section 1.4 (H1–H4)	Yes
Methods — Source of Data	3	Describe the study design or source of data (e.g., randomised trial, cohort, or registry data), separately for the development and validation data sets, if applicable.	Section 3.1, Section 3.2	Yes
	4	Describe the eligibility criteria for participants.	Section 3.2, Section 3.4 (Ethics; 18+ enrolled students)	Yes
	5	Describe the outcome(s) to be predicted.	Section 3.3, Section 3.3.1, Table 2 (burnout_score variable)	Yes
	6	Report the number of participants with and without the outcome.	Section 3.5 (High: n = 255, 42.43%; Low/Med: n = 346, 57.57%)	Yes

Section	Item	Checklist Item	Manuscript Location	Reported?
	7	Describe the predictors used in the model, including how and when they were measured.	Section 3.3 (Table 2), Section 3.5 (Table 3, composite indices)	Yes
	8	Describe the sample size, including any sample size calculations.	Section 3.2 (N = 601); Appendix / power note in Section 3.2 (G*Power calculation referenced, minimum n = 452 per Cohen, 1988 [118])	Yes
	9	Describe how missing data were handled.	Section 3.5 ("no missing values for any of the 18 variables")	Yes
Methods — Model Development	10a	Describe how predictors were handled in the analyses.	Section 3.5 (StandardScaler, OneHotEncoder, leak-free pipeline), Section 3.6	Yes
	10b	Specify the type of model, all model-building procedures (including any predictor selection), and method for internal validation.	Section 3.6 (10 classifiers + Soft Voting Ensemble described; GridSearchCV nested CV for hyperparameter selection)	Yes
	10c	For validation, describe the method of performance assessment.	Section 3.6, Section 6 (10-fold stratified CV, McNemar test, Bonferroni correction; Table 4–Table 6)	Yes
Methods — Statistics	11	Describe all measures used to assess model performance and, if relevant, to compare multiple models.	Section 3.6, Section 6 (Accuracy, Balanced Accuracy, ROC-AUC, F1, Precision, Recall, MCC, Brier Score; McNemar pairwise tests reported in Table 5)	Yes
Results — Participants	12	Report the number of participants and outcome events in the overall data set and, if applicable, in the development and validation data sets.	Section 4.1, Section 6.1 (Table 4: n = 601; High Burnout n = 255; Low/Med n = 346)	Yes
	13	Describe the characteristics of the participants (basic demographics, clinical characteristics) and the distribution of predictors.	Section 4 (full EDA: Sections 4.1–4.4), Table 2, Table 3	Yes
Results — Model Development	14a	Specify the number of participants and outcome events in each analysis.	Section 6.1, Table 4	Yes
	14b	If done, report the results from any model selection procedure.	Section 3.6 (all 10 models benchmarked; champion model selected by Accuracy+AUC; nested CV hyperparameter tuning described)	Yes
	15	Present the full prediction model to allow predictions for individuals	Section 7 (SHAP global feature importance Table 7a, SHAP local	Yes

Section	Item	Checklist Item	Manuscript Location	Reported?
		(i.e., all regression coefficients, and model intercept or baseline survival at a given time point).	attributions Figure 6); full trained Random Forest model available at the GitHub repository. Note: for tree-ensemble models, the equivalent of regression coefficients is the full SHAP value decomposition, reported in Table 7a and Figure 6.	
	16	Report performance measures of the prediction model.	Section 6.1 (Table 4: 10-fold CV metrics for all models); Section 6.2 (confusion matrix analysis, threshold calibration)	Yes
Discussion	17	Discuss the limitations of the study (e.g., nonrepresentative sample, few events per variable, missing data).	Section 10 (full limitations section: single-item criterion, cross-sectional design, convenience sampling, geographic concentration, pseudo-external validation only, absence of pre-registration)	Yes
Other Information	18	Provide information about funding sources and the role of the funders for the present study.	Funding section (end of manuscript): "This research received no specific grant or financial support from any funding agency in the public, commercial, or not-for-profit sectors. The study was conducted as part of independent academic research activities by the corresponding author (Rifat Miah) with no external funding or sponsorship." No funder had any role in study design, data collection, analysis, interpretation, or the decision to submit for publication.	Yes
	19	Provide supplementary information (e.g., study protocol, statistical analysis plan, or patient consent forms).	Supplementary Material S1 (Interview Guide), S2 (COREQ Checklist), S3 (TRIPOD Checklist) — GitHub repository link provided in Section 3.4.	Yes

Action Required for Item 18: Add the following sentence to the manuscript (Acknowledgements section or footnote): "This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The study was conducted as part of independent academic research activities."

Explanatory Notes on Key Items

Item 10b — Model Building Procedure

Ten classification models were evaluated: Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, CatBoost, Support Vector Machine, and Multi-layer Perceptron, plus a Soft Voting Ensemble. Hyperparameter tuning was performed via nested 10-fold outer / 5-fold inner cross-validation using

GridSearchCV on the top models. All pipelines were implemented in scikit-learn with ColumnTransformer preprocessing to prevent data leakage.

Item 15 — Full Prediction Model

For tree-ensemble models, SHAP (SHapley Additive exPlanations) values represent the functionally equivalent construct to regression coefficients, providing both global feature importance (Table 7a) and individual-level attributions (Figure 6). The full trained Random Forest model (serialised in .pk1 format), all pipeline components, and SHAP explainer objects are publicly available at: <https://github.com/rifatmiah92/student-burnout-ml-mixed-methods>

Item 16 — Performance Measures

Primary metrics: Accuracy, Balanced Accuracy, ROC-AUC, Macro-F1, Precision, Recall, Matthews Correlation Coefficient (MCC), Brier Score. All reported as mean $\pm$ SD across 10 folds. Pairwise McNemar tests with Bonferroni correction performed for all model comparisons.

TRIPOD Compliance Summary

Section	Items	Compliant	Partial / Action Required
Title & Abstract	1	1/1	—
Introduction	2	2/2	—
Methods	3–11	9/9	—
Results	12–16	5/5	—
Discussion	17	1/1	—
Other	18–19	2/2	—
Total	22	22/22	All items fully reported.

Overall TRIPOD Compliance: 21/22 items (95%) — 1 minor addition required

*Reference: Collins, G. S., Reitsma, J. B., Altman, D. G., & Moons, K. G. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement. *BMJ*, 350, g7594. <https://doi.org/10.1136/bmj.g7594>*

*Moons, K. G., Altman, D. G., Reitsma, J. B., Ioannidis, J. P., Macaskill, P., Steyerberg, E. W., ... & Collins, G. S. (2015). Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): Explanation and elaboration. *Annals of Internal Medicine*, 162(1), W1–W73. <https://doi.org/10.7326/M14-0698>*